

ABC Phones

Credit Portfolio: From Raw Snapshots to Reliable Insights

133%

Loan-book growth across 2025 (8.8K → 20.5K loans)
but credit quality deteriorating

44.8%

Latest delinquency rate — collection rate fell
10pp from Q1 to Q4 (69 → 60%)

+5pp

Default-rate gap for loans without complete
customer profiles vs the rest of the book

What we found in the data

Three sources, real-world inconsistencies, ~50% customer-side coverage gap

71,448

credit rows across 5 quarterly snapshots

20,742

unique loans (1 row per loan per snapshot)

3,985

NPS responses, 100% match credit

~50%

credit loans w/ no DOB/income/gender

Top quality issues identified

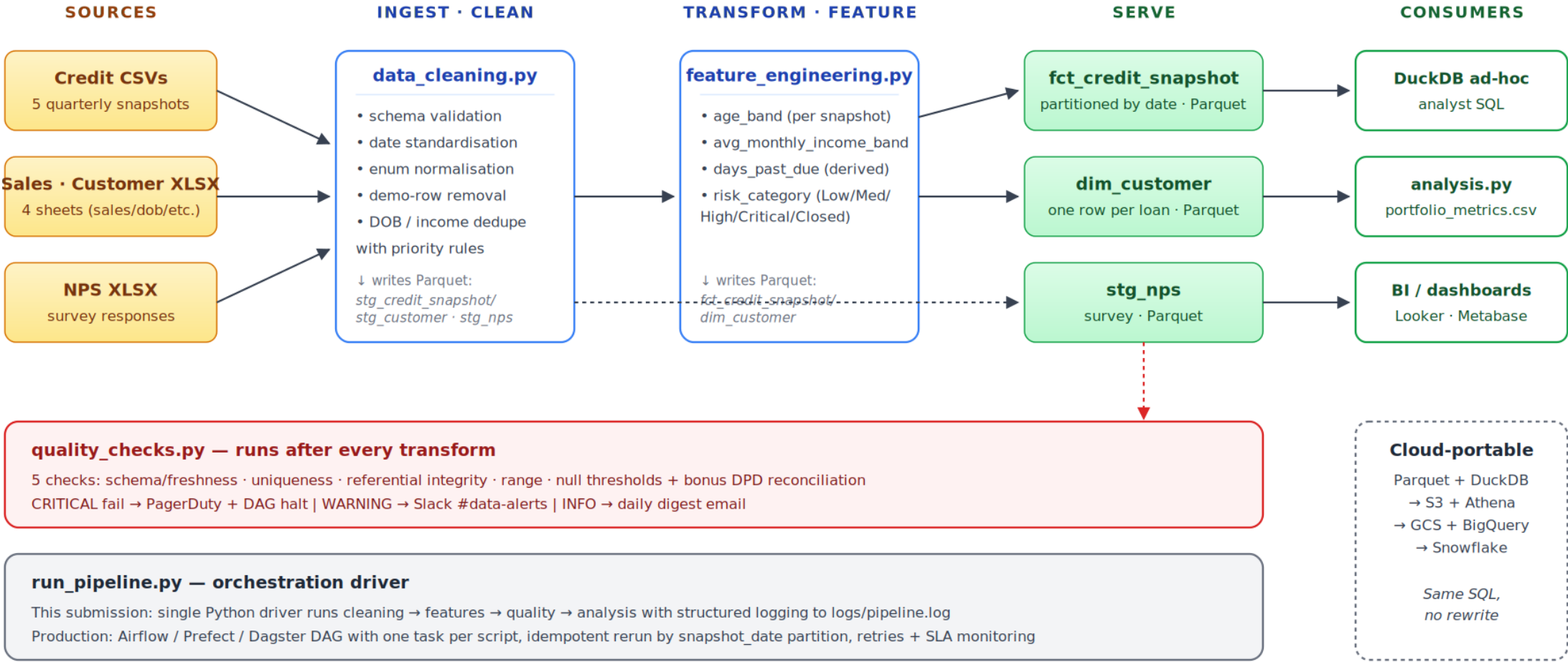
Issue	Detail	Resolution
Filename ↔ data drift	Credit_Data_-_30-03-2025.csv contains DATE = 3/31/2025	Caught by schema_and_freshness check (CRITICAL)
No CUSTOMER_ID anywhere	Joins are loan-to-loan only via LOAN_ID / Loan Id	Documented gap; multi-loan customers not trackable
Column-name typo	DOB sheet uses 'Loan Id ' (trailing space)	Stripped on ingest
DOB conflicts	471 loans have multiple DOBs across 3 providers	Resolution rule: TransUnion → SmileID → SpinMobile, then most recent createdAt
Income column ambiguity	4 income fields likely overlap — naive sum double-counts	Use 'Received' as canonical total; documented
Excel padding	All customer sheets padded to 1,048,575 rows (Excel cap)	dropna(how='all') on ingest
Inconsistent enum casing	BALANCE_DUE_STATUS: 'Arrears' vs 'up to date' vs 'advance'	Lowercased on ingest

ETL architecture

ABC Phones — Credit Portfolio ETL Pipeline

Python · Pandas · DuckDB · Parquet | Lakehouse pattern, cloud-portable

Lakehouse pattern on a laptop today, lifts to S3+Athena / GCS+BigQuery / Snowflake



Hive-partitioning by snapshot_date enables idempotent reruns and incremental loads — drop a partition, replace it, leave the rest untouched.

Every step is a pure function with a CleaningReport / CheckResult — easy to unit-test and trace.

Key engineering decisions

Each call is small but compounds into a robust pipeline

Hive partitioning by snapshot_date

Each snapshot is its own Parquet partition. Reruns drop & replace one folder; the rest stays untouched. Idempotent without locking.

DuckDB over Pandas-only

Cleaning stays in Pandas (row-wise logic). Analytics moves to SQL via DuckDB on the Parquet files. Same SQL ports to BigQuery/Snowflake.

Pure-function stages

Every `clean_*` and `check_*` function returns (df, report). No globals, no in-place mutation. Trivially unit-testable.

Source values preserved alongside derived

We keep both `days_past_due` (source) and `days_past_due_derived` (ours). The reconciliation gap itself is a quality check — not a bug to hide.

Schema contract on ingest

Expected columns are codified in `EXPECTED_CREDIT_COLS`. Missing or extra cols halt the pipeline before bad data reaches analysts.

Late-arriving / reprocessing strategy

Replay any snapshot independently — ``python run_pipeline.py --only cleaning`` rebuilds one partition. Nothing else recomputes.

Data quality framework

Five checks + one bonus, each with severity, cadence, and a real alerting route

Check	Severity	Cadence	Alert route
1. Schema & freshness — files arrive, columns match contract, internal DATE = filename	CRITICAL	real-time	PagerDuty + halt DAG
2. Uniqueness of (loan_id, snapshot_date)	CRITICAL	daily	PagerDuty + halt DAG
3. Referential integrity — every credit/NPS loan_id exists in dim_customer	WARNING	daily	Slack #data-alerts
4. Range checks — age 18-100, income > 0, NPS 0-10, DPD ≥ 0	WARNING	daily	Slack #data-alerts
5. Null thresholds — critical fields < 1%, soft fields < 60%	WARNING	daily	Slack #data-alerts
+ DPD reconciliation — derived vs source within 7-day tolerance	INFO	weekly	Daily digest email

REAL EXAMPLE FROM THE DATA

Filename ↔ internal date mismatch

Credit_Data_-_30-03-2025.csv contains an internal DATE column with value **3/31/2025** — silent date drift that would land in PAR-30 quarter-end metrics.

→ Caught automatically by check #1 with severity=CRITICAL on first run. Alert routed to PagerDuty; DAG halted before downstream tasks ran.

Feature engineering

Four derived fields per the brief; risk_category is where the judgement shows

Four required features

age_band

from DOB at each reporting date — 18-25, 26-35, 36-45, 46-55, 55+, plus explicit 'Unknown' bucket so coverage stays visible

avg_monthly_income_band

received_total ÷ duration_months → 8 bands per brief. 'Received' used as canonical total to avoid sub-channel double-counting

days_past_due

(reporting_date – next_invoice_date), clamped to 0 when no arrears. Source value retained for reconciliation

risk_category

the credit-risk judgement call → see right

risk_category logic

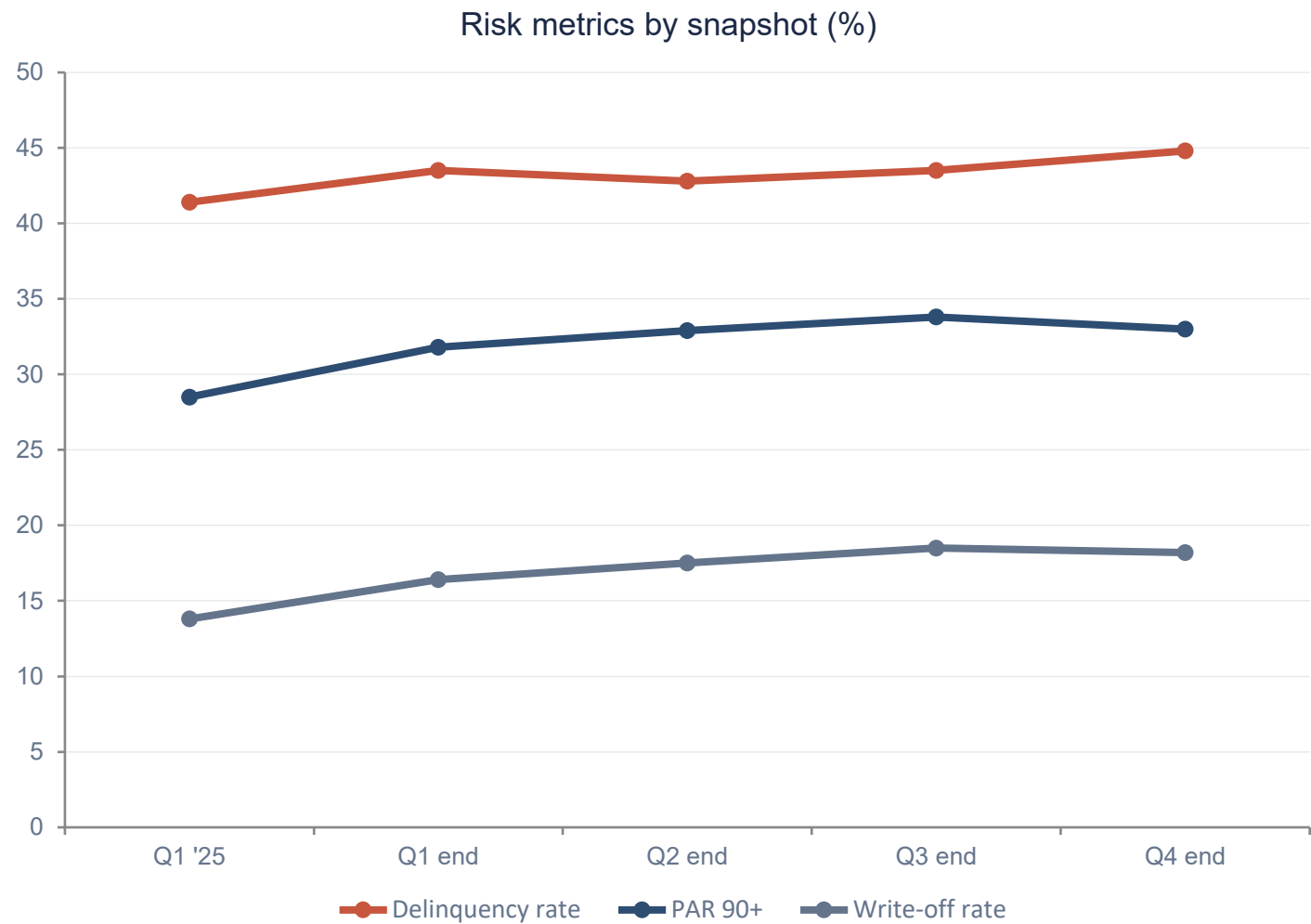
Standard PAR-bucket framework adapted to ABC Phones' status taxonomy

CRITICAL	Write Off / Lost Write Off OR DPD ≥ 90
HIGH	DPD 31-89 OR L2 ∈ {PAR 30, FMD}
MEDIUM	DPD 1-30 OR L2 ∈ {PAR 7, FPD, Inactive}
LOW	DPD = 0 AND L2 = Active
CLOSED	L2 = Paid Off (out of active book)
RETURNED	L2 = Return (out of active book)

Mirrors IFRS-9 staging logic and standard microfinance PAR-bucketing.

Portfolio health: book grew fast, quality slipped

Five quarterly snapshots, 2025-01-01 through 2025-12-30



59.5%

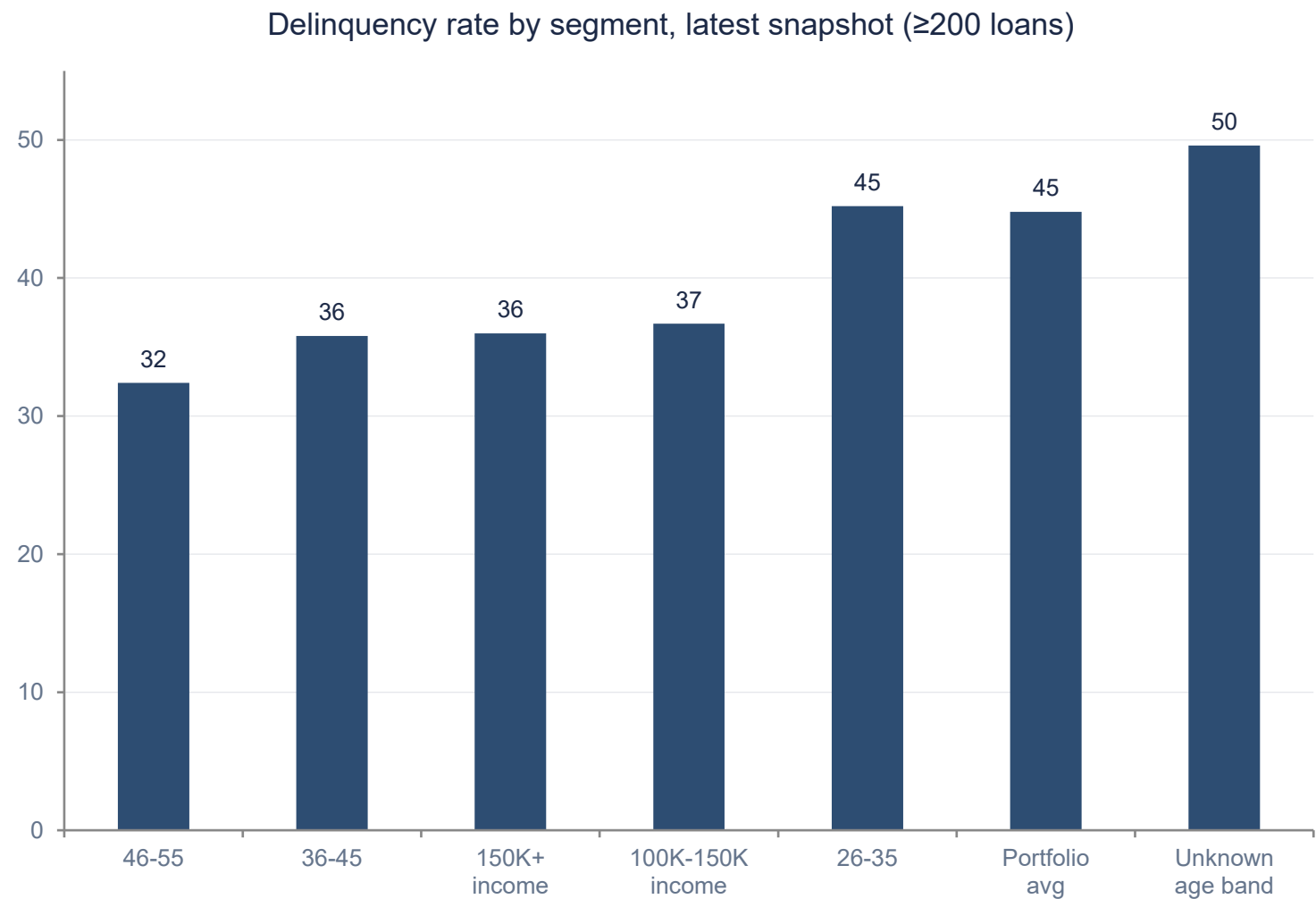
Active-loan collection rate (latest) — down from 69.4% Q1

Take-aways

- Book size more than doubled in 2025 (133% growth in active loans).
- Delinquency, PAR 90, and write-off rates all climbed in parallel — not a one-off.
- Collection rate fell 10 percentage points — the leading indicator.
- Pattern is consistent with rapid origination outpacing underwriting controls.

Where the risk really lives

Loans without complete customer profiles default ~5pp more than the rest of the book



THE FINDING

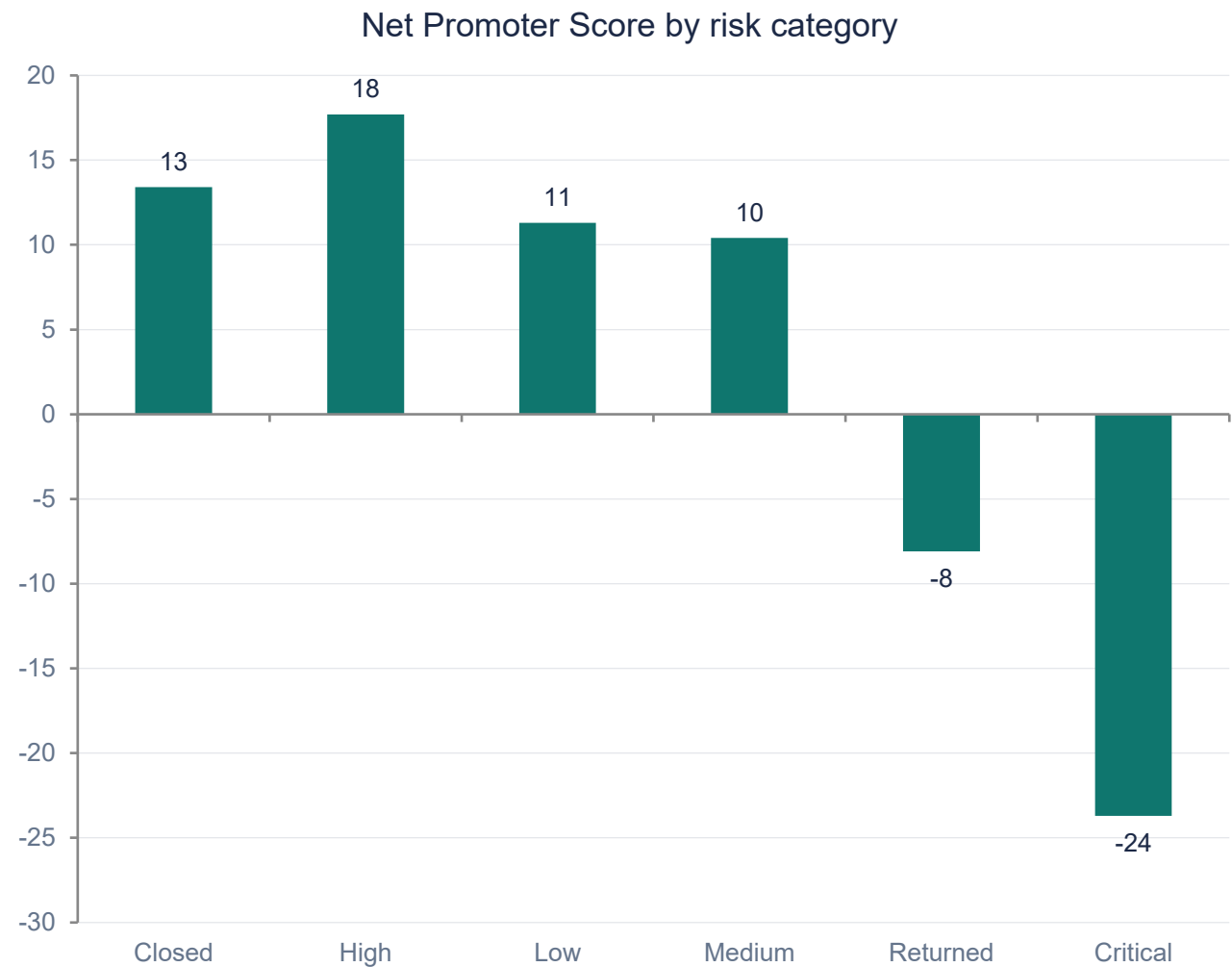
Loans with no DOB on file (46% of latest book) default at 49.6% — almost 5pp above portfolio average.

Loans WITH complete profiles (older borrowers, higher income) consistently perform 8-12pp better.

This isn't only a data gap — it's a credit-decisioning gap. The pipeline approves loans without complete profiles, and those loans default more.

Credit × NPS — and one fix that helps both

Each NPS response joined to the customer's most recent credit snapshot



RECOMMENDATION

Fix the lock-after-payment issue

12-17% of NPS respondents across every risk segment report *'phone locked despite paying on time'*

This drives **both** outcomes the business cares about: customer dissatisfaction (Critical-risk NPS = -23.7) AND avoidable arrears flags when the locking system runs ahead of payment reflection.

Fix: a 24-hour grace window before remote-locking, plus a reconciliation job that lifts locks within 1 hour of payment confirmation.

→ improves NPS and reduces 'fake' delinquency simultaneously — a rare win-win in collections design.

Where this goes next

Three data improvements ABC Phones should make + the production-roadmap delta

DATA IMPROVEMENTS

1

Introduce a stable `customer_id`

Hash of national ID or phone, propagated from KYC into credit and NPS. Unlocks lifetime-value, repeat-borrower risk, and household-level analysis.

2

Standardise on ISO-8601 + a YAML schema spec

Versioned alongside the data; documents the unit of every numeric column (Duration in months, balance in KES, etc.). Eliminates ambiguity discussed in Slide 2.

3

Replace daily snapshots with an event stream

Emit `payment_made`, `status_changed`, `arrears_updated` events; derive snapshots downstream. Removes the filename-vs-internal-date drift class of bugs entirely.

PRODUCTION ROADMAP DELTA

Orchestration

`run_pipeline.py` → Airflow / Prefect / Dagster DAG; one task per stage; SLAs

Storage

Local Parquet → S3 / GCS, same Hive partitioning

Compute

DuckDB → Athena or BigQuery — same SQL, no rewrite

Transforms

Python + SQL → dbt (lineage, docs, tests for free)

Quality

Custom Python checks → Great Expectations / dbt tests / Soda

Alerting

Mocked log routes → PagerDuty + Slack SDKs, on-call rota

Lineage

Add → OpenLineage events from each task